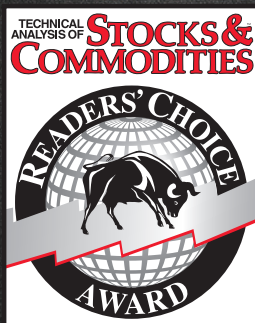


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Cycle Analysis:

A comparison of the Fourier and Maximum Entropy methods

by John F. Ehlers

The motivation for reducing price history to a mathematical expression is clear. If we can describe the prices mathematically, we have the means to extend the equation into the future to predict what the prices will be. Cycle analysis is particularly attractive because, if a simple cycle is present and will extend into the future, all we have to do is to wait until the prices reach a valley to buy and then wait again for the price to reach the crest of a cycle before we sell.

One way to represent prices mathematically is to describe the curve in terms of a power series expansion. The price as a function of time is written as the infinite series:

$$(1) P(t) = C_0 + C_1t + C_2t^2 + C_3t^3 + C_4t^4 + \dots$$

where the c_n 's are the coefficients of the respective n th power term. The first coefficient c_0 is just the average value of the price. The second term is linearly related to time, and therefore can be described as the trendline. The remaining terms introduce curves and the coefficients are adjusted to best approximate the price curve when they are summed together.

A power series is only one way to mathematically describe price, but it is perhaps the easiest to understand because one can relate it to the average price and the trendline. For this article, we want to describe the price in terms of sine waves to recover cyclic content.

Sine wave properties

We have seen that we can synthesize waveshapes as combinations of sine waves of different amplitudes ("Understanding Cycles," *Stocks & Commodities*, December, 1985). If we can synthesize waveshapes with sine waves, we can also analyze these waveshapes to recover each of the components which comprise the total. More generally, we can analyze any waveshape in terms of sine wave components. To do this, we represent the price as the doubly infinite 1 series:

$$(2) P(t) = a_0 + a_1\cos(x) + a_2\cos(x/2) + a_3\cos(x/3) + a_4\cos(x/4) + \dots + b_1\sin(x) + b_2\sin(x/2) + b_3\sin(x/3) + b_4\sin(x/4) + \dots$$

where x is shorthand for $2\pi t$

The number for each term is its period. For example, if the number of the period is 14 and time (t) increases from zero, then one cycle is complete when it reaches 14 because the phase of the sine wave would have reached 2π , or 360 degrees. As t is increased further, the second cycle of this term would

begin. The cycles can continue indefinitely. The relative amplitude of the a and b coefficients of the same number in the above equation determine the relative phase of the sine wave.

In the above equation, imagine that all the coefficients were zero except a_0 and a_{14} . This would be an ideal situation for a trader because the price description would consist only of an average value and a pure 14-day cycle. The purpose of cycle analysis is to recover the cycles when the situation is far more complicated.

If we can describe prices mathematically, we can extend the equations into the future to predict what the prices will be.

The physical picture of what we are trying to do is to run the data through a 2-day filter and measure the amplitude of the 2-day cycle, run the data through a 3-day filter and measure the amplitude of the 3-day cycle, and so on. We can essentially do this operation mathematically because sine waves are orthonormal (2) functions. The following properties are important for this procedure:

$$\int_0^{2\pi} \sin(mX)\sin(nX)dX = 0 \text{ for } m \neq n$$

$$\int_0^{2\pi} \cos(mX)\cos(nX)dX = 0 \text{ for } m \neq n$$

$$\int_0^{2\pi} \sin(mX)\cos(nX)dX = 0 \text{ for } m \neq n$$

$$\int_0^{2\pi} \sin(nX)\sin(nX)dX = \pi$$

$$\int_0^{2\pi} \cos(nX)\cos(nX)dX = \pi$$

These equations express properties which make mathematical analysis easy. For example, all we have to do to recover the amplitude of the 14-day cycle in equation 1 is to multiply both sides of the equation by $\cos(x/14)$ and take the integral. All terms on the right-hand side of the equation go to zero except the a_{14} term which is multiplied by π . Equation 2 then becomes:

$$\int_0^{2\pi} P(t)\cos(x/14)dx = \pi a_{14}$$

Therefore, the amplitude of the 14-day cycle coefficient is calculated by the evaluation of a simple integral.

Fourier analysis

Conventional Fourier analysis follows the above procedure to recover the amplitude of the coefficients of the Fourier series expansion of the function. Relative phase is lost in the process of taking the integral and so the power (3) in each component is found by adding the squares of the corresponding a and b

coefficients.

The main constraint of Fourier analysis is that the properties of the orthonormal functions require that the integral be taken over an integer number of cycles. This restricts the resolution available by this analysis. For example, assume we use 64 data points for analysis. Then, a 64-day cycle is the longest that can be found because longer cycles cannot complete a full cycle. In addition, the next shortest cycle that can be found is a 32-day cycle because it has two full cycles over the data length. The next shortest cycle available is $64/3=21.3$ days. Continuing, the cycles available are $64/4=16$ days, $64/5=12.8$ days, $64/6=10.7$ days, etc. There is no way to find the cycle amplitude for cycles between 16 days and 21.3 days except by increasing the length of the database to something greater than 64 days. This is a distinct disadvantage for short-term trading.

One of the problems with a longer database is that the phase of the shorter cycles can shift over the length of the database, canceling out its amplitude as calculated. However, this calculation can also be in error because the current value of the shorter cycle may be large. Perhaps saying the same thing another way is that the longer databases allow fundamentals to influence the short-term technical analysis. This lack of resolution is the fundamental drawback of conventional Fourier analysis for trading.

It perhaps needs mentioning that the database cannot be adjusted, to produce an integer number of cycles for each of the components because the implicit assumption is that the selected length of the data is repeated over and over in both time directions to infinity. If the database is adjusted for calculation, it amounts to changing the data and gross calculation distortions will occur.

Fast Fourier transforms

Longer databases are required to produce high resolution. Longer databases introduce a second problem with Fourier analysis. This second problem is that a larger number of calculations is required, and a large amount of computer time is consumed to accomplish the calculations. The problem is particularly acute with the interpreted BASIC language because the SIN and COS operations are performed in slowly accessed ROM lookup tables.

Algorithms have been developed that dramatically reduce the computation time for Fourier analysis. These are called Fast Fourier Transforms, or FFTs for short. One such program, written in Applesoft BASIC and suitable for trading analysis, is given in the listing.

Maximum Entropy method

Although the end result of finding the cycle amplitudes is the same, Maximum Entropy analysis is fundamentally different from Fourier analysis. In thermodynamics, entropy is a measure of unavailable energy. More appropriately, in information theory, entropy is the lack of information or the amount of noise. The analysis procedure is to remove cyclic content and then examine the residual for informational content. The process is repeated iteratively until the residual has maximum entropy.

No assumptions are made regarding the data outside the selected analysis period. That is, there is no constraint that the data repeat in either direction to infinity as was assumed with Fourier analysis. For this reason, there are no "end effect" distortions and distortions caused by the observation window are eliminated. Maximum Entropy responds maximally in the least-squared error sense to the data that it is presented and maximally ignores implied data.

The practical advantage of the Maximum Entropy method for trading is that a very high resolution can be

obtained from a very short database. Typically, the only data required is about one cycle's worth of information. For example, four weeks of data is entirely adequate to isolate a 17-day cycle with high fidelity. This short data requirement means that the data is fresh and relevant to the current trading picture. Since the data is fresh, there is a higher probability that a measured dominant cycle will extend into the future retaining its current phase.

Although the end result of finding the cycle amplitudes is the same, Maximum Entropy analysis is fundamentally different from Fourier analysis.

Retention of the phase relationship of the cycles is another advantage of the Maximum Entropy method. When the phase relationships are known, the cycles with their relative amplitudes can be extended into the future to form a complex prediction of the future prices. Under the right conditions this can be so accurate that it is almost scary.

I have written a computer program called MESA (for Maximum Entropy Spectral Analysis). An example of the program outputs is shown in Figure 1 for December, 1984 corn. The first panel is the price history with an optimized (half dominant cycle) moving average superimposed on the price. The suggested trading signals are generated by the price crossing the optimized moving average. The second panel shows that the dominant cycle is 10 days for the 20 days of data used in the analysis. (MESA automatically selects the proper data length.) A lower amplitude 6-day cycle has been identified by the program. In the bottom panel, the most recent 3-week price history is shown for continuity with the 3-week prediction of future prices.

Cycle analysis programs can be evaluated by synthesizing waveshapes from various sine waves and seeing if the program can take the composite wave shape apart to reproduce its components. Figure 2 shows such an analysis for MESA. I have synthesized a sawtooth waveshape from a dominant 30-day cycle, a 15-day cycle (second harmonic) at half amplitude, and a 10-day cycle (third harmonic) at one-third amplitude. Several periods of the sawtooth are shown in the top panel along with the optimized moving average.

The spectral components analyzed by the program are shown in the second panel. The 30-day dominant cycle was identified as a 28-day cycle, a 6.7 percent error. Still, this was done by using only 35 days of data. Moreover, the periods of the 15-day cycle and 10-day cycle were identified exactly. The half amplitude of the 15-day cycle and 10-day cycle was correctly identified at -6 dB, but the amplitude of the 10-day cycle should have been about -10 dB. I think the prediction in the bottom panel is awesome! The timing of the peak and the following valley of the predicted sawtooth is exactly the same as if the sawtooth were really allowed to extend into the future.

Conclusions

Cycle analysis can be helpful for trading in markets that are non-trending. One can even identify simple cycles by "eyeball," and can use the existence of the cycle to advantage with a variety of technical indicators. Fourier analysis requires a relatively long database to produce results with high resolution (good selectivity). On the other hand, the Maximum Entropy method requires only a very short database to produce high resolution, with the result that the data is current and pertinent to the anticipated trades.

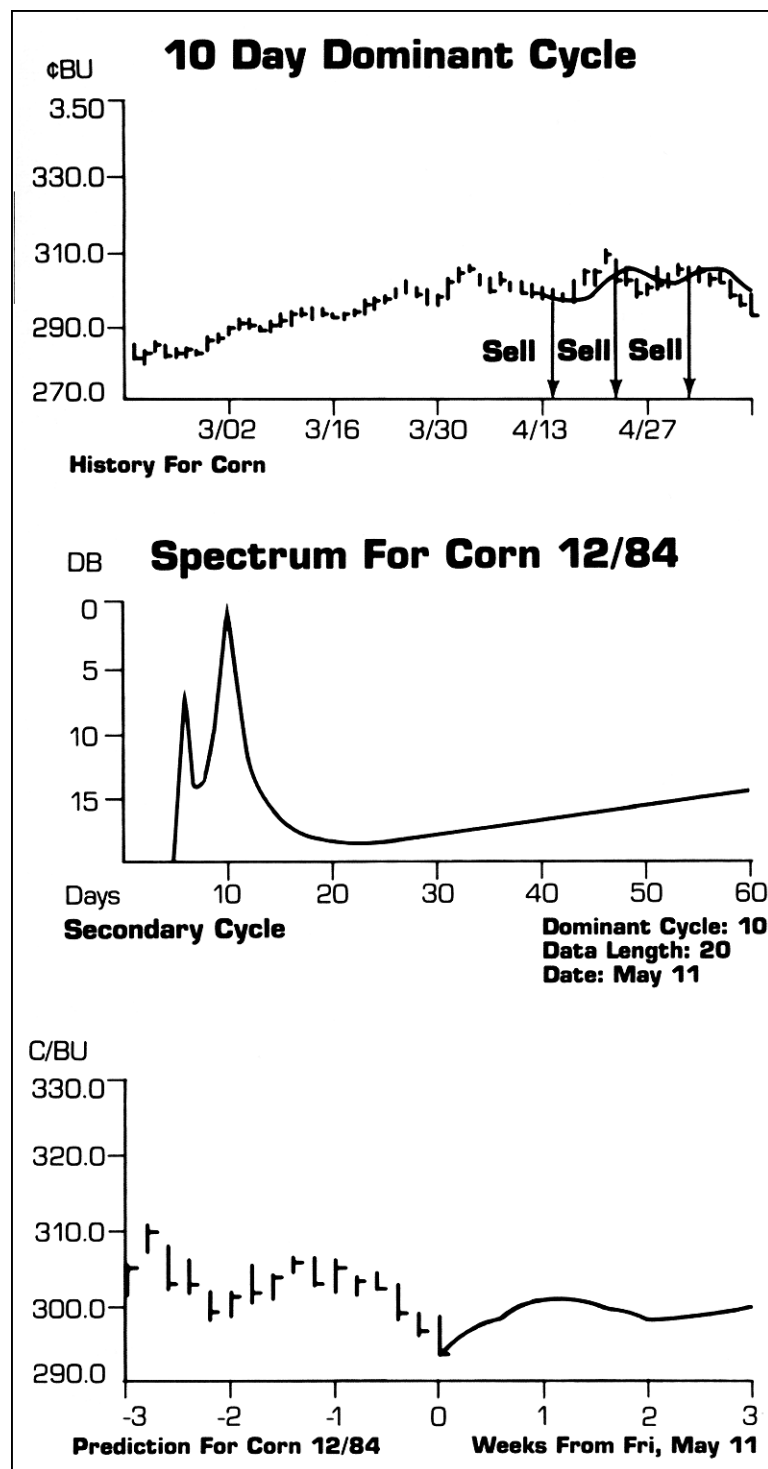


FIGURE 1

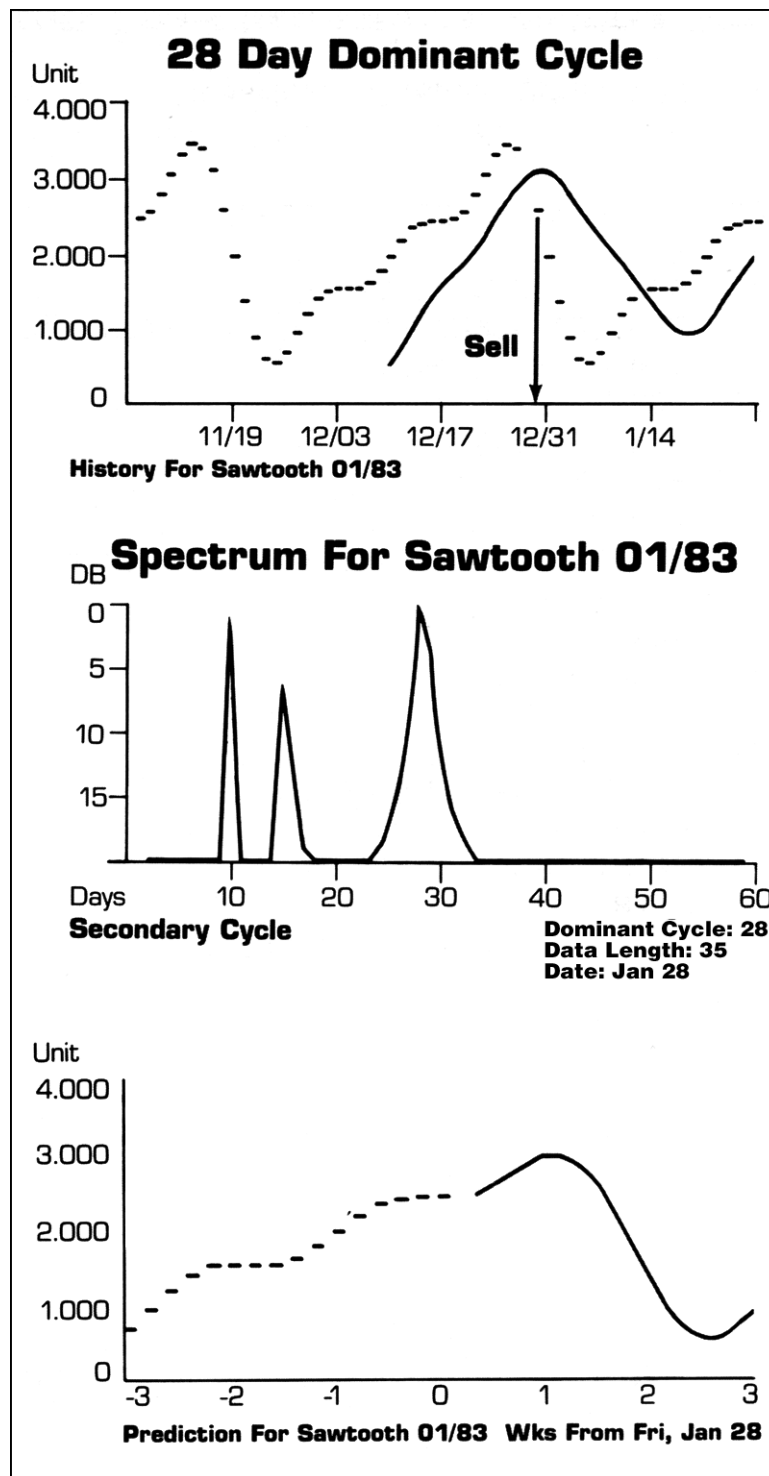


FIGURE 2

Maximum Entropy does not do an outstanding job on accurately measuring the relative amplitudes of the cycles and can be prone to producing spurious results, particularly if the database is too long. A significant feature of the Maximum Entropy method, complementing the use of current data, is that the relative phases of the cycles are retained so that a composite prediction of the future prices can be formed on the basis of the recent price history.

Glossary:

(1) Doubly infinite series- There is an infinite series of cosine terms and an infinite series of sine terms in the expression of equation 2.

(2) Orthonormal- A set of real and continuous functions is said to be orthonormal in the range (a,b) if

$$\int_a^b U_m(t)U_n(t)dt = \begin{cases} 1 & \text{if } m = n \text{ (normal condition)} \\ 0 & \text{if } m \neq n \text{ (orthogonal condition)} \end{cases}$$

When the functions satisfy the first condition, they are said to be "normal"; when they satisfy the second condition, they are said to be "orthogonal." The normal condition means the amplitude is normalized over a given integration range, and allows the coefficients of the functions to be evaluated. The orthogonal condition means that the solution of an expression can be accomplished by lineally adding the different functions because they do not interact over the integration period for which the coefficients are evaluated. Sine and cosine functions of the Fourier transform are only one kind of orthonormal functions. Other examples include Bessel functions, LaGuerre polynomials, Legendre functions, and Hermitian polynomials. These others have little application to trading.

(3) Power- I use the word "power" through an analogy to electrical waves. That is, each of the cosine waves and sine waves can be viewed as voltages. Electrical power is proportional to the square of voltage. The evaluation of the two lower equations on page 20 can thus be viewed as the square of the voltage terms (taken over the cycle), and are therefore analogous to power.

Fast Fourier Transform Program

```

10 REM "FAST FOURIER TRANSFORM"
20 REM FOR APPLE II WITH 2 DISKS USING
30 REM DATA IN CSI OR COMPU-TRAC FORM
40 REM BY JOHN F. EHLERS
45 REM COPYRIGHT (C) 1986 BY TECHNICAL ANALYSIS, INC..
50 HOME
60 DIM DF$(20)
70 HTAB 5:
  INVERSE :
  PRINT "*" FAST FOURIER TRANSFORM
  NORMAL
71 HTAB 19:
  PRINT "BY"
72 HTAB 14:
  PRINT "JOHN EHLERS"
73 HTAB 16:
  PRINT "BOX 1801"
74 HTAB 12:
  PRINT "GOLETA, CA 93116":
  PRINT
80 LET D$ = CHR$(4)
90 PRINT D$ + "OPEN MASTER,L40,D2"
100 FOR I = 1 TO 20
110   PRINT D$ + "READ MASTER,R";I
120   INPUT DF$(I)
130   IF LEFT$(DF$(I),5) = "99999"
140     GOTO 150
150 NEXT I
160 PRINT D$ + "CLOSE"
170 FOR J = 1 TO J - 1
180   PRINT "<" CHR$(J + 64); "> ";
190   MID$(DF$(J),4,16)
200 NEXT J
210 PRINT "(<V> VIEW NEW DATA DISK"
220 PRINT "(<X> EXIT TO MENU":
230 PRINT
240 PRINT "SELECT (< >)"; CHR$(8);
250   CHR$(8);
260 POKE - 16368,0:
270 GET X$:
280 PRINT X$;" "; CHR$(8); CHR$(8);
290 IF X$ = "V" THEN
300   HOME :
310   VTAB 10:
320   PRINT "INSERT NEW DATA DISK IN
330     DRIVE 2":
340   PRINT :
350   PRINT "PRESS ";
360   INVERSE :
370   PRINT "RETURN";
380   NORMAL :
390   PRINT "TO CONTINUE":
400   VTAB 22:
410   POKE - 16368,0:
420   GET S$:
430   HOME :
440   IF X$ = "X" THEN
450     HOME
460   LET SF = ASC(X$) - 64
470   IF SF >= 1 AND SF <= J - 1
480     THEN
490       290
500   HTAB 13:
510   PRINT " "; CHR$(8);
520   GOTO 220
530   IF K < 120 OR K > LR THEN
540     PRINT "DOPS -RUN AGAIN":
550     END
560   HOME :
570   VTAB 10:
580   INVERSE :
590   PRINT MID$(DF$(SF),4,16):
600   NORMAL :
610   PRINT :
620   PRINT
630   PRINT :
640   PRINT D$ + "OPEN " + MID$(DF$(SF),4,16) + ",L40":
650   PRINT D$ + "READ " + MID$(DF$(SF),4,16) + ",R0":
660   INPUT X$:
670   LET LR = VAL(X$)
680   PRINT D$ + "CLOSE"
690   IF LR < 32 THEN
700     370
710   IF LR < 1024 THEN
720     410
730   PRINT "CONTAINS ";LR;" RECORDS"
740   PRINT "A MAXIMUM 1024 RECORDS ARE
750     ALLOWED":
760   PRINT
770   PRINT "PRESS ";
780   INVERSE :
790   PRINT "RETURN";
800   NORMAL :
810   PRINT "TO CONTINUE":
820   VTAB 22
830   PRINT "CONTAINS ONLY ";LR;" RECORDS"
840   PRINT
850   PRINT "AT LEAST 32 RECORDS ARE REQUIRED":
860   PRINT
870   PRINT "PRESS ";
880   INVERSE :
890   PRINT "RETURN";
900   NORMAL :
910   PRINT "TO CONTINUE":
920   VTAB 22
930   POKE - 16368,0:
940   GET S$:
950   HOME :

```

```

CLEAR :
GOTO 20
410 PRINT "CONTAINS ";LR;" RECORDS":
PRINT
420 PRINT "ENTER ONLY 32, 64, 128, OR
256"
430 PRINT "FOR ANALYSIS. THIS NUMBER
MUST BE"
440 PRINT "LESS THAN THE FILE LENGTH."
:
PRINT
450 INPUT "RECORDS FOR ANALYSIS? ";N
460 IF (N = 32 OR N = 64 OR N = 128
OR N = 256 OR N = 512) AND N < LR
THEN
480
470 PRINT :
PRINT "OOPS - RUN PROGRAM AGAIN":
END
480 DIM DA$(1024),F(1024),A(257,8)
490 PRINT :
PRINT D$ + "OPEN " + MID$(DF$(SF
),4,16) + ",L40"
500 FOR I = 1 TO N:
PRINT D$ + "READ " + MID$(DF
$(SF),4,16) + ",R";(LR - N + I
):
INPUT DA$(I):
NEXT I
510 PRINT D$ + "CLOSE"
520 REM *** DATA CONVERTER ***
530 IF MID$(DA$(1),23,5) < > "99999
" THEN
550
540 LET DA$(1) = DA$(2)
550 FOR I = 1 TO N
560 LET F(I) = VAL ( MID$( DA$(I)
,23,5))
570 IF MID$( DA$(1),23,5) = "9999
9" THEN
LET F(I) = F(I - 1)
580 NEXT I
590 REM *** SPECTRUM ANALYSIS ***
600 LET PI = 3.1415926
610 LET R0 = INT (N / 4):
LET S0 = 8
620 FOR I = 1 TO R0 + 1:
FOR J = 1 TO 8:
LET A(I,J) = 0:
NEXT J:
NEXT I
630 FOR I = 1 TO R0
640 FOR J = 1 TO S0 STEP 2
650 LET K = 4 * (I - 1) + .5
* (J + 1)
660 LET A(I,J) = F(K)
670 LET A(I,J + 1) = 0
680 NEXT J
690 NEXT I
700 LET M = LOG (N) / LOG (2):
LET N2 = N / 2:
LET N1 = N - 1:
LET J = 1
710 FOR I = 1 TO N1
720 IF I >= J THEN
850
730 LET R = INT ((J - 1) / 4)
740 LET S = 2 * J - 8 * R
750 LET R1 = INT ((I - 1) / 4)
760 LET S1 = 2 * I - 8 * R1
770 LET R1 = R1 + 1
780 LET R = R + 1
790 LET T = A(R,S)
800 LET A(R,S) = A(R1,S1)
810 LET A(R1,S1) = T
820 LET T = A(R,S - 1)
830 LET A(R,S - 1) = A(R1,S1 - 1)
840 LET A(R1,S1 - 1) = T
850 LET K = N2
860 IF K >= J THEN
910
870 LET J = J - K
880 LET K = K / 2
890 GOTO B60
900 NEXT I
910 LET J = J + K
920 NEXT I
930 LET L0 = 1
940 FOR L = 1 TO M
950 LET L1 = L0
960 LET L0 = 2 * L0
970 LET V = 1
980 LET W = 0
990 LET Z = PI / L1
1000 LET W1 = COS (Z)
1010 LET W2 = SIN (Z)
1020 FOR J = 1 TO L1
1030 FOR I = J TO N STEP L0
1040 LET K = 1 + L1
1050 LET R1 = INT ((K - 1)
/ 4)
1060 LET S1 = 2 * K - 8 * R1
1070 LET R1 = R1 + 1
1080 LET A1 = A(R1,S1 - 1)
1090 LET B1 = A(R1,S1)
1100 LET T = A1 * V - B1 * W
1110 LET U = A1 * W + B1 * V
1120 LET R = INT ((I - 1)
/ 4)
1130 LET S = 2 * I - 8 * R

```

```

1140      LET R = R + 1
1150      LET A(R,S1 - 1) = A(R,
S - 1) - T
1160      LET A(R1,S1) = A(R,S)
- U
1170      LET A(R,S - 1) = A(R,S
- 1) + T
1180      LET A(R,S) = A(R,S) + U
1190      NEXT I
1200      LET U = V * W1 - W * W2
1210      LET W = V * W2 + W * W1
1220      LET V = U
1230      NEXT J
1240      NEXT L
1250      LET Z = - 1E6
1260      FOR I = 1 TO R0 / 2
1270          FOR J = 1 TO S0 STEP 2
1280              IF I = 1 AND J = 1 THEN
1310                  LET A(I,J) = SQR (A(I,J)
* A(I,J) + A(I,J + 1) * A(
I,J + 1))
1300              IF A(I,J) > Z THEN
LET Z = A(I,J)
1310          NEXT J
1320      NEXT I
1330      HOME :
PRINT "SPECTRUM FOR " + MID$(DF$(
SF),4,16)
1340      PRINT
1350      PRINT "PERIOD      RELATIVE AMPLITUD
E"
1360      PRINT "(DAYS)      (DB)":
PRINT
1370      LET K = 0
1380      LET P = 1000
1390      FOR I = 1 TO R0 / 2
1400          FOR J = 1 TO S0 STEP 2
1410              IF I = 1 AND J = 1 THEN
1460                  LET L = INT (N / K + .25)
1430                  IF L + 1 > P THEN
1460                      LET P = L
1450                      PRINT TAB( 2);L; TAB( 16);
INT (1000 * LOG (A(I,J)
/ Z) / LOG (10)) / 100
1460                      LET K = K + 1
1470              NEXT J
1480      NEXT I
PROGRAM LENGTH: 152 LINES / 3304 BYTES

```




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